CL205 2026 **Project Problem Statement**

Please use Python to solve the question below

Due Date: 24th August 2026, Midnight

Q1. This question involves using simple and multiple linear regression on the Auto data set to build the best model predicting mpg from the various response variables available in the dataset.

a. Use mpg as a predictor and each of the other variables (horsepower, number of cylinders etc) as a response variable, compute linear regression coefficients for one response variable at a time.
b. Use the python notebook uploaded on moodle as a template to compute the regression coefficients etc from scratch.
c. Using the Auto dataset, perform a bootstrap analysis to estimate the sampling distribution of the least-squares regression coefficient for horsepower when predicting mpg. Generate a histogram of the bootstrap slope estimates and mark the 2.5th and 97.5th percentiles.
d. Use the resulting bootstrap distribution to construct a 95% confidence interval for the regression coefficients and interpret the relationship between horsepower and mpg.
e. Compute R^2 for each linear regression fit.
f. Which variable is the best predictor?
g. Now compare a linear and quadratic regression model with mpg as the response and each of the response variables as the predictor. Hint (for quadratic regression with horsepower as the response variable compute mpg = f(horsepower, horsepower^2))
h. Build multiple regression models combining all the variables in linear and quadratic form. Find the best possible model.

Q2. We will now look at this as a classification problem using logistic regression.

a) Create a binary variable, mpg01, that contains a 1 if mpg contains a value above its median, and a 0 if mpg contains a value below its median. You can

compute the median using the median() method of the data frame. Note you may find it helpful to add a column mpg01 to the data frame by assignment.

b) Explore the data graphically in order to investigate the association between mpg01 and the other features. Which of the other features seem most likely to be useful in predicting mpg01? Scatterplots and boxplots may be useful tools to answer this question. Describe your findings.

c) Split the data into a training set and a test set.

d) Perform logistic regression on the training data in order to predict mpg01 using the variables that seemed most associated with mpg01 in (b). What is the test error of the model obtained?